1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L9 – Case Study (Other Functional Requirements)**

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4 **Recap**

- We are discussing a case study for the "requirement gathering, analysis & specification" stage
  - Learned about usability requirement gathering
- Today – functional requirements for the same case study (calendar app)

5 **Example Scenario**

- A calendar app
  - We are interested to build a calendar app. It is meant for the students (mainly) to help in their various academic activities
  - Let's try to find out the requirements to build this app (and also specify those)

6 **Functional Requirements**

7 **Functional Requirements**

- Recap - main idea
  - —Abstraction - identify black box functions to be supported by the system (with clearly defined input and output)
  - — Decomposition & hierarchy - to better manage the functional description

8 **Requirements at a Glance!**

- The very first thing we should consider - how to display dates
  - There are many possibilities
  - Should fixate on one (and specify)

9 **Requirements at a Glance!**

- Next – support for a student's academic activities
  - Setting reminder for lectures
  - Setting reminder for exams
  - Setting other academic event reminders (peer meetings, talks …)

We shall use the term "event" to refer to all the above (lectures/exams/meetings/talks…)

10 **Requirements at a Glance!**

- Also – some additional functionalities
  - Synchronize with email accounts (so that any extrinsic event/meeting created with the email account is also reflected on the calendar with date and time)
  - Set/reset background

11 **Specify Requirements**

- Using SRS notation
  - Create function name with descriptions
  - Specify input and output of the functions

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SRS – Calendar App

13 **Requirement for "Calendar View"**

- For a user to view dates with events (if any)
  - Requirement to view "months" for a year
  - Requirement to view "days" for a month
  - Requirement to view "event listings" for "days"
  - Also, requirement to "jump" to "any date" with events listed for that day

14 **Display Calendar - A Hierarchical Depiction**

15 **SRS – Calendar View (Top Level)**

R1: Display Calendar
Input: Year/Month/Date
Output: Months/Days/Event Lists
Description: In this function, the user can view months of a specified year or days of a specified month. The user can also view the events/deadlines in a month or days of a month or on a specific day.

16 **SRS – Month View (Hierarchy)**

R1.1: Display Month
Input: Year
Output: Months of the year
Description: In this function, the user can view months of a specified year.

17 **SRS – Day View (Hierarchy)**

R1.2: Display Days with Events
Input: Month
Output: Days of the month, along with day-wise event lists (if any)
Description: In this function, the user can view the days of a specified month. For each day, the associate event list (if any) is also displayed.

18 **SRS – Date View (Hierarchy)**

R1.3: Specific Date View
Input: Date
Output: Events
Description: In this function, the user can view the events for a specific date.

19 **Requirement for "Event Reminder Setting"**

- For a user to set reminders for academic events day wise
  - Requirement to "create" day-wise event(s)
  - Requirement to "edit" day-wise event(s)
  - Requirement to "delete" day-wise event(s)

20 **Reminder Setting - A Hierarchical Depiction**

21 **SRS – Reminder Setting (Top Level)**

R2: Set Reminders
Input: Date and event details with time
Output: Reminder for the event at the specified time on the specific date
Description: In this function, the user can create/edit/delete reminders for specific academic events on specific dates and times.

22 **SRS – Create Reminder (Hierarchy)**

R2.1: Create Reminder
Input: Date, event name, time
Output: Reminder for specific event
Description: In this function, the user can associate a specific date and time with a specific academic event.

23 **SRS – Edit Reminder (Hierarchy)**

R2.2: Edit Reminder
Input: Date, reminder
Output: Date with revised reminder
Description: In this function, the user can modify specific reminder(s) (such as time or event name) for specific date(s).

24 **SRS – Delete Reminder (Hierarchy)**

R2.3: Delete Reminder
Input: Specific reminder
Output: Date without the specified reminder
Description: In this function, the user can remove specific reminder(S) associated with specific date(s).

25 **Some Other Requirements**

- To "set" background
  - Requirement to "add" background
  - Requirement to "reset" background

26 **Hierarchical Depiction**

27 **SRS – Set Background**

R3: Set Background

Input: Images
Output: App background with chosen image
Description: In this function, the user can set/reset app background.

28 **SRS – Set Theme**

R3.1: Set Theme
Input: Theme image
Output: App background with chosen image
Description: In this function, the user can set app background.

29 **SRS – Reset Theme**

R3.2: Reset Theme
Input: Theme image
Output: App background with chosen image
Description: In this function, the user can reset app background.

30 **Some Other Requirements**

- To "synchronize" with other e-mail accounts
  - Requirement to "add" account to the app
  - Requirement to "delete" account from the app

31 **Hierarchical Depiction**

32 **SRS – Synchronize**

R4: Synchronize
Input: Details of account to synchronize with
Output: Synchronized reminders
Description: In this function, the user can synchronize the calendar with his/her respective email-id(s) so that any extrinsic event/meeting created with the email account is also reflected on the calendar with date and time

33 **SRS – Synchronize**

R4.1: Attach Account
Input: Details of account(s) to synchronize with
Output: Synchronized reminders
Description: In this function, the user can attach email-id(s) to the app for synchronization.

34 **SRS – Synchronize**

R4.2: Remove Account
Input: Details of account(s) to un-synchronize
Output: Un-synchronized reminder list
Description: In this function, the user can remove synchronized account(s) from the app.

35 **Complete FR Hierarchy**

36 **Note**

- How functions are represented (name, input, output, description)
- Abstract (black-box) nature of the functions – what we need, NOT how to get it!

- Hierarchical organization
- Naming convention

37 **Note**

- Next, we shall explore inclusion and specification of the usability requirements in the SRS